"The Phenomenon of Embodiment: The Evolution of Ontology from Properties to Interfaces"

(An Axiomatic Treatise)

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Based on the MPO-System (Method of Properties — Ontological Operating System)

Introduction for the Reader Unfamiliar with the MPO-System

This An Treatise is devoted to a systematic investigation of the **Phenomenon of Embodiment** — a fundamental process underlying the entire dynamics of reality. Here, "embodiment" is understood not merely as the transition of an idea into a material object (as in everyday language), but as a universal mechanism by which one and the same structure — be it a mathematical relation, an emotional image, or a physical pattern — can manifest across qualitatively distinct ontological media: in the world of matter, in the world of meanings, in the world of consciousness, and beyond.

The work is grounded in the author's ontological system — **MPO-System (Method of Properties — Ontological Operating System)** — and develops the concept of **Superreality**: a unified field of absolute connectivity in which an infinite multitude of autonomous, mutually irreducible ontological regimes ("worlds") coexist. In contrast to

traditional metaphysics, which proceeds from the primacy of matter, spirit, or information, the MPO-System renounces any monism and instead offers a **non-reductionist, pluralistic architecture of being**, wherein no single level of reality is primary.

The central result of the analysis is the introduction of **Property 38 — "Interfaceness"**, which captures the ontological status of a boundary not as a passive divider, but as an active zone of translation, mutual determination, and the generation of novelty. This property makes it possible to rigorously describe how transitions between worlds occur without violating their autonomy and without resorting to energetic or informational metaphors.

The monograph is intended for readers interested in formal ontology, the philosophy of mind, the theory of complex systems, and the foundations of artificial intelligence. Prior familiarity with the MPO-System is not required — all key concepts are introduced and explained within the text.

Aim, Methodology, and Structure of the Work

The aim of this investigation is to provide a systematic account of the **Phenomenon of Embodiment (henceforth PE)** as a universal processual mechanism through which Superreality enacts its own dynamics. Achieving this aim requires solving several interconnected tasks: (1) to explicate the ontological foundation — the concept of Superreality and the axiomatics of the MPO-System; (2) to define PE, its structure, and its key features; (3) to introduce and justify a new property — interfaceness; (4) to analyze the interrelations between PE and the axioms and basic properties of the system; (5) to demonstrate the empirical significance of the model, including its implications for understanding evolution, consciousness, and technological progress; (6) to compare the obtained results with alternative ontological projects.

Methodologically, the work relies on **operational phenomenology** — an approach in which investigation is directed not toward the search for "hidden entities," but toward the fixation and analysis of discernible structures of transition between ontological regimes. Phenomena are regarded as acts of embodiment unfolding within the field of connectiv-

ity, and theoretical constructs are introduced as needed and "dismantled" at the earliest opportunity (like scaffolding) to avoid hypostatizing intermediate concepts. This method allows the inquiry to remain within a rigorous phenomenological stance, without slipping into either physicalism or speculative metaphysics.

The structure of the monograph follows the logic of unfolding the problem. **Part I** establishes the ontological foundation: the concept of Superreality and the four axioms of the MPO-System are introduced, forming the axiomatic core of the entire construction. **Part II** is devoted to defining the Phenomenon of Embodiment, analyzing the structure of the act of embodiment, and its key features. **Part III** introduces Property 38 — interfaceness — and examines the world of boundaries (W_4) as an autonomous ontological regime. **Part IV** reveals the mutual constitution of PE and the axiomatics, showing how each axiom conditions PE and simultaneously acquires operational expression within it. **Part V** presents a detailed analysis of interfaceness: its formal definition, internal structure, relation to other MPO-System properties, and the role of auto-interfaces in intra-world dynamics. **Part VI** investigates the connection between PE and scale asymmetry and derives an ontological model of evolution that does not require a Darwinian mechanism. **Part VII** outlines the empirical implications of the model and possible practical applications. **Part VIII** contains a comparative analysis of the MPO-System with the process philosophy of Whitehead, the actor-network theory of Latour, and the object-oriented ontology of Harman. Finally, **Part IX** summarizes the results of the investigation, formulates philosophical implications, and charts directions for further research. The monograph is supplemented by appendices: an operational schema of PE, a glossary of terms, and a map of ontological worlds.

Part I. The Ontological Foundation: Superreality and the Axiomatics of the MPO-System

Chapter 1. The Concept of Superreality

Defining Superreality as a Unified Field of Absolute Connectivity

Superreality is the fundamental concept of the MPO-System, designating a unified, dynamic **field of absolute connectivity** in which all ontological regimes (worlds) and all entities are actualized. Unlike traditional ontological categories such as "being," "substance," or "nature," Superreality is neither an aggregate of objects, nor a container, nor a generative principle. It is the very field of connectivity that precedes any distinctions into objects and relations. Objects, properties, worlds — all are secondary stabilizations arising as local configurations of this field.

This understanding rests on the axiom of absolute connectivity (**KSS** $\rightarrow \infty$, see Chapter 2), which asserts that all entities across all conceivable ontological regimes are interconnected. Connectivity here does not imply physical interaction or the exchange of energy; it is a purely ontological characteristic expressing the fact that no entity exists in isolation. Even what is perceived in everyday experience as an independent object is in fact a node within an infinite network of connections permeating all worlds.

Superreality, therefore, is not something "external" with respect to worlds or entities. It is immanent to them, constituting the very fabric of their existence. Worlds (**ChOR**) are different regimes of actualization of this connectivity, distinguished by their "ontological flesh" (substrate), internal laws, and types of entities. Transition between worlds is possible only through the Phenomenon of Embodiment, which, as will be shown, is precisely the operational enactment of connectivity.

Rejecting Monism and Hierarchical Models of Being

Traditional metaphysics has, as a rule, striven toward monism — the reduction of the entire manifold of existence to a single first principle: matter, spirit, idea, will, informa-

tion, etc. Even in those systems that acknowledged a plurality of levels of being (e.g., in Plato or Neoplatonism), these levels were arranged in a strict hierarchy, where the higher was regarded as more genuine and causally determinative with respect to the lower. The MPO-System breaks radically with both stances.

First, the **rejection of monism** means that Superreality is not reducible to any one of its ontological regimes. Matter is not more primary than consciousness, nor consciousness more primary than matter; meaning is not an epiphenomenon of the physical, nor is the physical "congealed" meaning. Each world (W_1 — the material, W_2 — the semantic, W_3 — the phenomenal, etc.) possesses its own ontological autonomy and is not derivable from other worlds. Attempts at reduction — be they physicalism, idealism, or panpsychism — are declared category mistakes arising from a confusion of ontological regimes.

Second, the **rejection of hierarchy** means that worlds are not nested within one another like Russian dolls and do not form a ladder of perfection. They are **juxtaposed**. The semantic world is not "above" the physical, nor is the phenomenal world "between" them. Connection between worlds is accomplished not through emanation or subordination, but through **interfaces** — acts of embodiment occurring in a special world of boundaries (W_4). Such pluralism excludes both reductionism and the notion of "higher" and "lower" levels of reality, replacing them with a topology of equipollent yet qualitatively distinct regimes.

It is precisely this dual rejection — of monism and of hierarchy — that clears the space for the introduction of the *Phenomenon of Embodiment* as the sole mechanism of connection between ontologically equipollent yet qualitatively distinct worlds.

The Principle of Ontological Pluralism: An Infinite Multitude of Irreducible Regimes

The principle of ontological pluralism is formalized in the axiom **ChOR** $\to \infty$ (see Chapter 2), according to which Superreality consists of an infinite number of ontological regimes. These regimes, or worlds, are not variations of one and the same thing (like parallel universes in physical cosmology); they differ **qualitatively**, in the very type of

their existence.

Each world is characterized by three main aspects:

1. **Ontological flesh (medium)** — the specific "material" of which the entities of that world are "made." For W_1 it is mass-energy and fields; for W_2 — meanings, signs, logical relations; for W_3 — experiences, qualia, emotional states; for W_4 — events of transition, Γ-operators.

2. **Internal laws** — the regularities to which entities within that world conform. In W_1 these are physical laws; in W_2 — logical and grammatical rules; in W_3 — psychological and phenomenal regularities.

3. **Types of entities** — categories of objects, processes, and states possible within that world.

It is crucial to stress that worlds are **neither derivable from nor reducible to one another**. A mathematical theorem (W_2) does not consist of atoms and is not generated by neural processes, although its comprehension by a subject (W_3) may correlate with brain activity (W_1). Correlation is not identity or causality: distinct worlds are connected not causally, but through structural isomorphism established in acts of embodiment.

The infinity of the multitude of worlds (ChOR $\to \infty$) is of fundamental importance. It guarantees that chains of re-embodiment will never close within a finite set of media and that for any pattern there will always be an authentic medium into which it can be embodied. Moreover, the process of embodiment is itself capable of **generating new worlds**: thus, the world of boundaries (W_4) arises as a necessary condition for transitions between already existing worlds, and the world of virtual constructs (W_6) arises as a result of the embodiment of semantic structures in a digital medium. Hence, the ontological pluralism of the MPO-System is not static but dynamic: the multitude of worlds is open and capable of expanding.

Chapter 2. The Axiomatics of the MPO-System

The axiomatic core of the MPO-System is formed by four postulates that specify the fundamental properties of Superreality and determine the conditions of possibility for the Phenomenon of Embodiment. These axioms are not derivable from one another and are not reducible to more fundamental principles; they constitute the minimal set of ontological commitments necessary for constructing a non-contradictory picture of reality as a field of connectivity.

ChOR $\to \infty$ — Infinite Stratification of Ontological Regimes (Worlds)

The axiom of infinite stratification asserts: Superreality consists of an **infinite number of ontological regimes** (ChOR — *ontological regimes*), each of which possesses its own organization, laws, ontological flesh, and types of entities. Worlds are not nested within one another and are not derivable one from another.

The notation **"ChOR $\to \infty$"** captures two key points: (1) stratification — the presence of qualitatively distinct layers of being, irreducible to a common denominator; (2) infinity — the absence of an upper bound on the number of such layers. This does not mean that all possible worlds are actually given in experience; some are hypothetical or not yet generated. Yet the very structure of Superreality is such that it imposes no principled restrictions on the diversity of ontological regimes.

Within the MPO-System, the following basic worlds are distinguished:

- W_1 — **The Material World**: the regime of physical objects, fields, energy, space-time.
- W_2 — **The Semantic World**: the regime of meanings, signs, symbols, mathematical and logical structures.
- W_3 — **The Phenomenal World**: the regime of subjective experience, sensations, emotions, qualia.
- W_4 — **The World of Boundaries**: the regime of processes of transition, interfaces, and Γ-operators.

In addition, hypothetical worlds are considered: the normative (W_5), virtual constructs (W_6), chaotic patterns (W_7), and others. The list remains open in principle.

The axiom ChOR $\to \infty$ plays a critical role for the Phenomenon of Embodiment. Without a plurality of worlds, PE would be limited to transitions within a single medium and could not generate radically new types of entities. The infinity of worlds guarantees that the chain of re-embodiments will never hit a "final" regime and that for any pattern there is a medium capable of receiving its embodiment.

KSS $\to \infty$ — Absolute Connectivity of All Entities and Worlds

The axiom of absolute connectivity (**KSS** — *Cohesive Synergy Index*) asserts: **all entities across all ontological regimes are absolutely interconnected**. There are no isolated objects, impermeable boundaries, or ontologically autonomous monads. Connectivity is a fundamental characteristic of Superreality, not an epiphenomenon of interactions.

It is essential to distinguish **connectivity** from **interaction**. In the material world (W_1), interaction presupposes the exchange of energy or matter through local contact. Connectivity in the sense of KSS $\to \infty$ requires neither locality nor exchange. It means that between any two entities there exists an ontological channel that can be actualized in an act of embodiment. This is not a causal connection in the classical sense, but the possibility of establishing an isomorphism.

The axiom KSS $\to \infty$ is a necessary condition for the Phenomenon of Embodiment. Were the worlds ontologically isolated, the transition of a pattern from one world to another would be impossible. It is precisely KSS $\to \infty$ that guarantees that for any pair (source-world, target-world) there exists a Γ-operator capable of effecting the embodiment. In turn, PE renders connectivity operational: without acts of embodiment, connectivity would remain pure potentiality, an abstract possibility never passing into actuality.

PPU $\to \infty$ — Paradoxical Stability as a Condition for the Coexistence of Contradictions

The axiom of paradoxial stability (**PPU** — *Paradoxical Persistence Unit*) asserts: **Superreality is stable under contradictions, paradoxes, and incompatible descriptions**. The coexistence of mutually exclusive properties, laws, or descriptions does not lead to the collapse of the system; on the contrary, paradoxes are sustained as a productive tension that generates complexity and dynamics.

In classical logic and most formal systems, the principle of explosion (*ex contradictione quodlibet*) holds: from a contradiction anything follows, rendering the system trivial. **PPU** $\to \infty$ rejects this principle at the ontological level. Contradiction does not destroy Superreality; it becomes a source of new distinctions and transitions.

For the Phenomenon of Embodiment, the axiom PPU $\to \infty$ is of decisive importance. The act of embodiment is **paradoxical** in its very essence: a pattern, in passing from one medium to another, **remains the same** (preserves its invariant structure) and simultaneously **becomes other** (acquires new properties, conforms to new laws). Without PPU $\to \infty$, this paradox could not be stably realized; every act of embodiment would lead to logical collapse. PPU $\to \infty$ ensures that identity and difference can coexist without annihilating one another.

Moreover, PPU $\to \infty$ makes possible the coexistence of the worlds themselves. Different ontological regimes have different, often incompatible, laws. In the material world, the law of energy conservation holds; in the semantic world, meanings can multiply without energy expenditure; in the phenomenal world, experiences are not subject to physical metrics. PPU $\to \infty$ allows these regimes to coexist within a unified Superreality, without being reduced to one another and without coming into conflict.

SA $\to \infty$ — Scale Asymmetry: A Lower Bound on Distinction, No Upper Bound on Complexification

The axiom of scale asymmetry (**SA** — *Scale Asymmetry*) asserts: **the scale of distinction in Superreality is asymmetric**. In moving "inward" (decreasing scale), there

is a finite limit beyond which further distinction loses meaning. In moving "outward" (increasing scale), there is no limit — chains of embodiment can complexify without end.

Formally, **SA** $\to \infty$ captures two interconnected claims:

1. **Lower bound on distinction.** In each ontological world, there exists a minimal scale at which it is still meaningful to speak of "embodiment" and "distinction." For W_1 this limit is the Planck length (and the associated Planck units); for W_2 — atomic concepts that cannot be further decomposed without loss of meaning (e.g., "unit," "point"); for W_3 — thresholds of sensitivity below which experience becomes indiscernible. This bound imparts **finitude** to any concrete act of embodiment: a pattern cannot be embodied "more finely" than this threshold.

2. **Absence of an upper bound.** In the opposite direction — toward enlargement, complexification, integration — there is no principled limit whatsoever. A pattern can be embodied in ever more complex structures: from quarks to atoms, from atoms to molecules, from molecules to cells, from cells to organisms, from organisms to societies, and so on. The chain of re-embodiments is **open upward**.

Scale asymmetry has fundamental consequences for the Phenomenon of Embodiment. It imparts **directedness** to chains of re-embodiment: downward movement is bounded, upward movement is not. This explains the **irreversibility** of embodiments: exact reverse embodiment is impossible because, in moving down the scale, information about structure inaccessible at that level of resolution is lost. Each embodiment is unique and historical.

Furthermore, SA $\to \infty$ applies to the world of boundaries (W_4) itself. There exists a **minimal Γ-operator** — a quantum of transition below which the distinction "embodied / not embodied" vanishes. But **there is no maximal Γ-operator**: arbitrarily complex, recursive, and multi-level interfaces are possible. This explains the infinite recursion of boundaries ($W_4 \to W_5 \to \ldots$) and the principled openness of the architecture of Superreality.

Finally, SA $\to \infty$ serves as the ontological foundation for directed evolution (see Part VI). Because upward movement on the scale of complexity has no limit, while downward

movement is bounded, chains of re-embodiment necessarily exhibit a tendency toward complexification — without invoking external teleological principles or Darwinian selection.

The four axioms — **ChOR** $\to \infty$**, KSS** $\to \infty$**, PPU** $\to \infty$**, SA** $\to \infty$ — form an indissoluble unity that defines the architecture of Superreality. ChOR $\to \infty$ supplies the ontological material — an infinite multitude of worlds. KSS $\to \infty$ guarantees their connectivity. PPU $\to \infty$ renders this connectivity stable in the face of paradoxes. SA $\to \infty$ imparts directedness and irreversibility to the dynamics. Together, they create the conditions in which the Phenomenon of Embodiment can unfold as a universal processual mechanism — the subject of the next part.

Part II. The Phenomenon of Embodiment: Definition and Mechanism

Chapter 3. Defining the Phenomenon of Embodiment

The Phenomenon of Embodiment as the Transition of an Invariant Pattern from One Ontological Medium to Another

The **Phenomenon of Embodiment (PE)** is the central processual mechanism of Superreality. In the most general terms, it is defined as the **transition of an invariant pattern from one ontological medium (source-world) to another ontological medium (target-world), as a result of which a new entity possessing stable properties emerges in the latter**. By a pattern is meant a stable configuration of connections — a structure, relation, idea, potential — existing in the source medium in a state of pure potentiality (**Propertylessness, P25**). The result of embodiment is an entity with the status of **Onticity (P33)**, fully belonging to the target-world and subject to its laws.

It is of fundamental importance that PE **does not involve any transfer of matter, energy, or information** from one world to another. The pattern does not "travel"

between worlds; it remains in its authentic medium. The connection between the pattern and the embodied entity is established not through transfer, but through **structural correspondence (isomorphism)**. In the target-world, under the action of a special operator — the **Γ-operator** — a new entity is configured whose internal structure of relations is isomorphic to the structure of the pattern in the source-world.

This definition extends the concept of embodiment far beyond its narrow, physicalist sense (the transition from a "disembodied idea" to a "material thing"). Within the MPO-System, embodiment can occur between any ontological regimes: from the semantic world to the physical (idea → artifact), from the physical to the phenomenal (event → experience), from the phenomenal to the semantic (emotion → concept), and so on. Moreover, as will be shown below, PE also takes place within a single world, where one pattern is embodied in another within the same ontological medium.

Rejecting the Transfer of Matter, Energy, or Information

Traditional conceptions of causality and interaction are built on the metaphor of **transfer**: body A transfers momentum to body B; a source emits energy absorbed by a receiver; a sender encodes information and transmits it over a communication channel. In all these cases, the existence of some carrier (matter, field, signal) is assumed, which moves in space-time from source to receiver. The Phenomenon of Embodiment in the MPO-System is fundamentally distinct from such a model.

First, worlds (ChOR) are ontologically heterogeneous and lack a common substrate that could serve as a carrier. There is no "substance of meaning" that flows from W_2 to W_1 when an artifact is created from a blueprint; there is no "field of experiences" transmitting an emotion from one subject to another. Attempting to think PE in categories of transfer immediately leads to a category mistake — a confusion of ontological media.

Second, even when transfer of matter or energy does take place (e.g., in physical interaction), it describes only an **intra-world** process, not a transition between worlds. The impact of a billiard ball transfers momentum to another ball within W_1 according to the laws of physics; yet if that same impact was initiated by the conscious intention of

a player, the connection between the intention (W_3) and the ball's motion (W_1) is not a transfer of energy — the intention contains no joules.

Third, the rejection of information transfer as an explanatory principle stems from the fact that the very category of information in the MPO-System is regarded as an **embodied pattern**, not as a universal substance. Information always exists within a concrete ontological medium (physical states, symbolic structures, conscious contents) and cannot be "extracted" from it in a pure form. What cybernetics calls information transfer is, in terms of PE, a sequence of embodiments of one and the same pattern across a chain of media (e.g., acoustic signal → neural activity → semantic content).

Thus, PE establishes a connection between worlds **without transfer** of any substrate whatsoever. The sole "bridge" is structural isomorphism — a relation of correspondence that is neither matter, nor energy, nor information. This relation is discovered after the fact, in comparing the pattern at the source and the entity at the target, but in itself it has no ontological status as an independent entity.

The Concept of "Ontological Flesh": The Specificity of the Medium for Each World (W_1–W_n)

Each ontological world is characterized by its own **ontological flesh** (or **medium**) — the specific "material" in which the entities of that world exist and of which they are "made." The term "flesh" is used metaphorically to stress that what is meant is not physical matter, but any substrate capable of bearing stable configurations and conforming to internal laws.

For the currently known worlds of the MPO-System, the specification of ontological flesh is as follows:

- W_1 **(Material World).** Flesh: mass-energy, fields, space-time. The entities of this world — particles, waves, macroscopic bodies, radiation — conform to the laws of physics (classical and quantum). Embodiment in W_1 means that a pattern acquires physical properties: mass, charge, spatial localization, temporal dynamics.

- W_2 **(Semantic World).** Flesh: meanings, signs, symbols, logical and mathematical

relations. The entities of W_2 — concepts, theorems, texts, linguistic constructions — have no mass and are not localized in physical space; their existence is determined by intrasystemic rules (logical consistency, semantic coherence, syntactic correctness).

- W_3 **(Phenomenal World).** Flesh: experiences, sensations, emotions, qualia. The entities of W_3 — pain, joy, a visual image, a conscious intention — are given directly to the subject and possess a qualitative determinacy (phenomenal character) irreducible to physical or semantic descriptions.

- W_4 **(World of Boundaries).** Flesh: events of transition, acts of embodiment, Γ-operators. The entities of W_4 are not stable objects but processual units: measurement, translation, insight, collision. They belong neither to the source nor to the target, but constitute the very boundary between worlds.

- W_5–W_n **(Hypothetical worlds).** For the normative world (W_5), the flesh consists of imperatives and norms; for the virtual (W_6) — code and memory states; for the chaotic (W_7) — dynamic instability and attractors. The list remains open.

The concept of ontological flesh performs a dual function in the MPO-System. On the one hand, it underscores the qualitative difference of worlds and the impossibility of their reduction: one cannot "make" a thought out of atoms or a physical object out of meanings. On the other hand, it indicates precisely **into what** a pattern is embodied in a transition. Embodiment is always concrete: one and the same pattern (e.g., the relation of symmetry) in W_1 becomes a crystal lattice, in W_2 — a mathematical group, in W_3 — an aesthetic experience of harmony. These embodiments are not identical to one another, yet they are structurally isomorphic, and it is precisely the ontological flesh that determines which concrete properties the result will acquire.

Chapter 4. The Structure of the Act of Embodiment

The act of embodiment unfolds as a sequence of three phases, unified by the action of the Γ-operator. This structure is invariant for all types of PE — both inter-world and intra-world. Each phase is examined in detail below.

Phase 1: The Initial State — The Pattern in a State of Propertylessness (P25)

Prior to the act of embodiment, there exists in the source-world a **pattern** — a stable configuration of connections possessing a definite structure. However, with respect to the target-world, this pattern is in a state of **Propertylessness (P25)**. This means that it **does not possess** those concrete properties characteristic of entities in the target-world.

For instance, the idea of a chair in the mind of a designer (W_3/W_2) has no mass, size, or color, and occupies no place in physical space. The mathematical concept of a circle (W_2) is not round in a physical sense. An emotional image (W_3) is not composed of neurons, although it correlates with definite neural activity (W_1). The pattern in P25 exists as **pure potentiality**: it can be embodied, but in itself it is not yet actualized in the target medium.

The state P25 is characterized by a **supersaturation** of possibilities. One and the same pattern is potentially capable of being embodied in different worlds, generating qualitatively distinct entities. Thus, the relation "greater–less" can be embodied in a physical hierarchy of sizes, in a logical order relation, in an emotional gradation of intensity. The concrete choice of medium and the concrete result of embodiment are determined in the next phase.

Phase 2: The Γ-Operator as an Event-Interface Actualizing Connectivity

The central element of the act of embodiment is the **Γ-operator** (gamma-operator). This is neither a force, nor a cause in the classical sense, nor an agent, but an **event-interface** occurring in the world of boundaries (W_4). The Γ-operator actualizes the connectivity pre-given by the axiom KSS $\to \infty$ and effects the transition of the pattern from state P25 to state P33.

Functionally, the Γ-operator acts as a **catalyst of recoding**. It "resonates" with the pattern in the source-world and, using the ontological material (flesh) available in the target-world, confers upon the pattern a stable, discernible form. In so doing, the Γ-operator does not "extract" the pattern from the source nor "insert" it into the target; it rather creates the conditions under which the target-medium spontaneously configures

itself in accordance with the structure of the pattern.

The Γ-operator always operates through **interfaceness (P38)** — the property of a boundary to be a zone of translation. The interface (W_4) is polyvalent: it can serve a multitude of source–target pairs and can vary in intensity — from weak (the boundary between oil and water) to extreme (the event horizon of a black hole, the moment of creative insight). The intensity of the interface correlates with **salience (P37)** — the ontological trigger that initiates the transition upon reaching a critical threshold.

It is important to stress that the Γ-operator is **not an elementary, indivisible event**. Even the simplest act of embodiment (e.g., the collision of billiard balls) unfolds as a cascade of micro-Γ-operators, forming a complex interfacial zone with its own internal dynamics. This inexhaustibility of the Γ-operator has fundamental epistemological significance, limiting the possibilities of a complete formalization of PE.

Phase 3: The Result — An Embodied Entity with the Status of Onticity (P33)

The result of the act of embodiment is an **embodied entity** that emerges in the target-world and possesses the status of **Onticity (P33)**. This means that it:

- stably exists in its authentic medium;
- possesses concrete, discernible properties conforming to the laws of that world;
- can enter into interactions with other entities of the same world;
- can itself become a pattern (P25) for subsequent acts of embodiment.

The embodied entity is **not identical** to the original pattern, yet it is **isomorphic** to it with respect to its core structure. It also exhibits **emergence (P4)** — it acquires new properties irreducible to either the properties of the pattern or those of the medium. For example, a physical computer (W_1) embodying the abstract idea of computation (W_2) possesses characteristics such as heat dissipation, clock frequency, and material consumption that were absent from the original pattern and do not follow logically from it.

The status P33 is **graded**: entities can possess varying degrees of ontological complexity, measured by **propertyness ($\mathcal{N}_p$)**. Primitive configurations (e.g., a crystal lattice) have low $\mathcal{N}_p$; highly organized systems (living organisms, consciousness, social institutions) have high $\mathcal{N}_p$. The growth of $\mathcal{N}_p$ in the chain of re-embodiments is one manifestation of scale asymmetry (SA $\to \infty$).

Isomorphism as the Basis of Connection Between Worlds

The connection between the pattern in the source and the embodied entity in the target is secured not by the transfer of any substrate, but by **structural isomorphism**. Isomorphism is a relation of one-to-one correspondence between the elements and connections of two structures such that their formal organization is preserved.

In the context of PE, isomorphism means that the core configuration of connections defining the pattern in the source-world is reproduced in the target-world by means of its ontological flesh. The concrete "bearers" of connections and the concrete "materials" may be entirely different. For example:

- The relation "being a part of" can be embodied in a physical composite structure (atom $\to$ molecule), in a logical hierarchy (genus $\to$ species), in a social organization (individual $\to$ group).
- The dynamic pattern "cycle" is embodied in the oscillations of a pendulum (W_1), in a logical circle (W_2), in a recurring experience (W_3).

Isomorphism **is not an independent entity** or a "third world." It is a **relation** that can be fixed by an observer in comparing source and result. In itself, it exerts no causal influence and requires no ontological carrier. In this sense, isomorphism is analogous to the mathematical notion of a morphism: it is a way of describing structural correspondence, not an object.

It is precisely isomorphism that permits us to speak of the "same" pattern embodied in different worlds while avoiding substantialist pitfalls. There is no "pure pattern" hovering above the worlds; there are concrete configurations in concrete media, and there

is a relation of structural similarity between them. The Phenomenon of Embodiment is the process of **establishing** such an isomorphism, translating potentiality into actuality without the transfer of any substrate.

Chapter 5. Key Features of the Phenomenon of Embodiment

Invariance of the Pattern Across Transitions

A fundamental characteristic of PE is the **invariance** of the core structure of the pattern across transitions between media. Despite the radical change of ontological flesh and the acquisition of new, context-specific properties, the pattern preserves its identity — that very configuration of connections that makes it "the same" across different embodiments.

Invariance does not mean absolute immutability. The pattern may be enriched, lose inessential details, be scaled, but its **core** — the minimal structure of relations necessary for recognition and functionality — remains stable. For example, the idea of a "wheel" preserves the invariant (round shape, rotation about an axis) when embodied in a wooden cart wheel, in the metal rim of a bicycle, in the abstract concept of a "cycle," or in the metaphor of the "wheel of samsara."

It is precisely invariance that secures the **transcendental unity** of chains of re-embodiment. Without it, each new embodiment would be absolutely new, unconnected to preceding ones, and the very idea of "embodiment" would lose its meaning. At the same time, invariance does not require the postulation of Platonic ideas or other eternal entities: it is an empirically fixable structural constancy, discovered in comparing different realizations.

Ontological "Flesh" as the Specificity of the Medium

A second key feature of PE is the decisive role of **ontological flesh** (the medium) in shaping the concrete properties of the embodied entity. If invariance is responsible for preserving the "form," the medium is responsible for the "matter" of embodiment.

In transitioning to a new medium, the pattern inevitably acquires properties charac-

teristic of that medium and conforms to its laws. A mathematical theorem embodied in a physical experiment acquires measurement errors, temporal duration, and material costs. An emotional experience embodied in a poetic text acquires rhythm, rhyme, and cultural connotations. A physical object embodied in a phenomenal image acquires subjective coloration, an affective tone, and mnemonic associations.

The medium **is not a passive "material"** out of which the Γ-operator "sculpts" a form. It actively participates in the process of embodiment, imposing constraints, offering possibilities, and ultimately co-determining the result. In this sense, PE is not a unilateral "influence" of the pattern upon the medium, but an **interaction** (within W_4) between structural potential and medial conditions, in which a new entity is born.

The specificity of the medium explains why one and the same invariant core can generate such diverse phenomena in different worlds, and why the reduction of one embodiment to another (e.g., consciousness to neurophysiology) is a category mistake: in transitioning between media, it is precisely that ontological flesh which constitutes the specificity of a given world that is lost.

Reversibility and Cyclicity: From "Embodiment" to "Re-Embodiment"

The term "embodiment" can create the illusion of a one-time, linear act: something "disembodied" becomes "corporeal," and the process is complete. However, analysis shows that any act of PE is a link in an **infinite chain of transitions**. The embodied entity (P33) immediately becomes a pattern (P25) for the next act of embodiment — in the same or a different medium. Hence, it is more accurate to speak not of "embodiment" but of **re-embodiment**.

The chain of re-embodiments **has neither beginning nor end**. The question "Which came first, the chicken or the egg?" is meaningless in this perspective: the "chicken" (embodiment of a biological pattern in an organism) and the "egg" (embodiment of the same pattern in a reproductive structure) always already coexist as two mutually presupposing links of a cycle. The attempt to find a "first" link presupposes linear causality, whereas PE describes a **cyclic network**.

Reversibility of PE holds in the sense that a transition from world A to world B is possible, as is the reverse — from B to A. A physical object can be embodied in an idea (perception → concept), and an idea can be embodied in a physical object (blueprint → artifact). Yet **exact reversibility is impossible** due to scale asymmetry (SA → ∞) and emergent enrichment. Reverse embodiment never reproduces the original pattern with absolute precision: it generates a new pattern, enriched by the experience of the forward transition. Therefore, the chain of re-embodiments does not close into identity but develops, generating unique entities at each step.

Totality: PE Operates Both Between Worlds and Within a Single World

Although the primary intuition of PE is linked to transitions between qualitatively distinct ontological regimes, the operation of this mechanism is not limited to inter-world transitions. PE is **total**: it takes place even within a single world, when one pattern is embodied in another within the bounds of the same ontological medium.

Examples of intra-world PE are numerous and span all known worlds:

- **In W_1:** the transfer of momentum in the collision of balls (a kinematic pattern is embodied in another body); electromagnetic induction (a magnetic pattern is embodied in an electric current); a phase transition (a chaotic pattern is embodied in a crystal lattice).

- **In W_2:** logical deduction (premises are embodied in a conclusion); metaphor (a literal meaning is embodied in a figurative one); the generation of a new concept (a combination of known meanings is embodied in a new concept).

- **In W_3:** association (one experience is embodied in another by similarity); reflection (a vague sensation is embodied in a conscious emotion); empathy (the experience of another is embodied in one's own feeling).

Intra-world PE is accomplished through **auto-interfaces** — boundaries between states of one and the same ontological medium. The auto-interface is a limiting case of interfaceness (P38), in which the inter-aspect (connection of different worlds) and the aut-

aspect (inner otherness) merge. Such transitions obey the same three-phase structure (P25 $\to \Gamma \to$ P33) and the same axioms as inter-world transitions.

The totality of PE has an important consequence: the dynamics of any world can be described in terms of the embodiment of patterns. Physical processes, semantic transformations, the stream of consciousness — all are chains of re-embodiments unfolding in their respective media. The unity of Superreality is secured not by substantial monism, but by the universality of the processual form of PE, which permeates all ontological regimes.

Thus, the Phenomenon of Embodiment appears as a universal mechanism for the actualization of the potential, operating at all levels of Superreality — from quantum events to creative insights, from billiard strikes to scientific revolutions. In the next part, the key property that makes this mechanism possible — interfaceness — will be introduced.

Part V. Interfaceness as a New Property of Ontology (Detailed Analysis)

Chapter 12. Property 38: Formal Introduction and Ontological Status

The Necessity of a New Property

Introducing a new property into an ontological system requires justification: why are the existing thirty-seven properties insufficient, and what lacuna is the thirty-eighth property intended to fill?

The analysis of the Phenomenon of Embodiment conducted in the preceding parts has revealed the fundamental role of the **boundary** as a zone of transition between ontological regimes. The Γ-operator, which effects the act of embodiment, always operates at the boundary of worlds — in a special zone that belongs neither to the source nor to the target. It is precisely in this zone that the translation of the pattern, the establishment of isomorphism, and the generation of a new entity take place. Yet none of the properties P1–P37 describes the ontological status of the boundary itself.

Consider adjacent properties that might lay claim to this role:

- **Property 34 "Bindability" (P34)** describes the capacity of a pattern and a medium to enter into a relation that generates an embodiment. It is an operator of transition, but not a characterization of the zone in which the transition occurs.
- **Property 11 "Non-Locality" (P11)** captures the possibility of connection without spatiotemporal contact, but does not specify the nature of the interface through which this connection is actualized.
- **Property 30 "Transformation" (P30)** describes a qualitative leap in the process of PE, but, again, not the locus of the leap itself.
- **Property 8 "Boundedness" (P8)** specifies the limits of an entity, but treats the boundary as a passive limit rather than an active zone of translation.

Thus, the traditional understanding of a boundary as a passive divider, line, or surface proves incompatible with the operational role that the boundary plays in acts of embodiment. To remedy this lacuna, **Property 38 — interfaceness** — is introduced.

Definition of Property 38 — "Interfaceness"

Interfaceness (P38) is the capacity of an ontological regime, state, or process to function as a **dynamic zone of translation, transformation, and mutual determination** between heterogeneous ontological media.

Formally, interfaceness can be characterized by the following theses:

1. Interfaceness **is not an attribute of an object**. It pertains not to "things" but to relations and events. An interface is not a thing, but a zone of contact and transition.
2. Interfaceness is **graded**. It manifests across a spectrum of intensities — from weak (minimal connectivity) to extreme (an event horizon, a moment of insight).
3. Interfaceness is **polyvalent**. One and the same interface can link not two, but a multitude of ontological regimes simultaneously.

4. Interfaceness is **constitutive for PE**. Every act of embodiment occurs through an interface; the Γ-operator is the actualization of interfaceness in a concrete event.

5. Interfaceness is **rooted in the axiom KSS** $\to \infty$. It is a local manifestation of absolute connectivity, the way in which the global network of relations becomes operationally accessible.

Within the context of the MPO-System, interfaceness acquires the status of a full-fledged, irreducible property. It is not derivable from other properties and is not their combination; it captures an independent aspect of the ontological architecture of Super-reality.

Distinction from Adjacent Notions

For a precise determination of the place of P38 within the system, it is necessary to demarcate it clearly from adjacent notions — both within the MPO-System and beyond it.

- **Distinction from a "boundary" in the topological sense.** In mathematics and physics, a boundary is defined as the set of points belonging to the closure of a set but not to its interior. This is a static, geometric notion. Interfaceness, by contrast, describes not a spatial limit but a **processual mode of the existence of connection**. An interface is not a line but an event.

- **Distinction from a "transmission channel" in information theory.** A channel presupposes the transfer of a signal from a source to a receiver through some medium. Interfaceness **does not presuppose the transfer of substrate**; it is the condition for establishing isomorphism without the transfer of matter, energy, or information. The Γ-operator does not "transmit" a pattern but creates the conditions for its reassembly in a new medium.

- **Distinction from "bindability" (P34).** Bindability describes the **possibility** of establishing a connection between a pattern and a medium — this is the operational

potential of the Γ-operator. Interfaceness describes the **ontological status of the very zone of connection** — this is not a possibility but a mode of existence of the boundary. One may say that P34 is a tactical mechanism, while P38 is a strategic structure: bindability is realized *within* the interface.

- **Distinction from the "world of boundaries" (W_4).** W_4 is an autonomous ontological regime whose "flesh" consists of events of transition. Interfaceness (P38) is a **property** that can manifest in various worlds (including W_4), characterizing concrete boundaries as zones of translation. W_4 is a medium; P38 is an attribute of a boundary within this (and not only this) medium.

- **Distinction from an "interface" in the technical sense.** In computer science, an interface is a specification of interaction between components (API, user interface). It is passive and auxiliary. Interfaceness in the MPO-System is an **ontological category**: the interface is active, it transforms structures passing through it, and it can itself generate new ontological regimes.

Thus, P38 fills a unique niche in the system of properties of the MPO-System, neither duplicating existing notions nor coming into conflict with them.

Chapter 13. The Structure of Interfaceness

Graded Intensity

Interfaceness is not a binary property that is either present or absent. It manifests across a **continuous spectrum of intensity**, from arbitrarily weak to extremely strong. This gradedness is not an epistemic artifact (a consequence of incomplete knowledge) but an ontological characteristic of the property itself.

The intensity of interfaceness correlates with two other properties of the MPO-System:

- **Salience (P37)** — the measure of transformative potential, the threshold of transition actualization.

- **Propertyness ($\mathcal{N}_p$)** — the quantitative measure of the ontological complexity of the embodied entity.

The higher the intensity of the interface, the more pronounced the moment of salience and the greater the $\mathcal{N}_p$ of the result. In limiting cases (e.g., the event horizon of a black hole), all three indicators — intensity of interfaceness, salience, and $\mathcal{N}_p$ — tend toward maximal values.

The spectrum of intensity can be heuristically divided into several characteristic regions:

- **Weak interfaceness.** The connection between media is minimal; the transition hardly alters the properties of the pattern. Examples: gravitational interaction between distant galaxies, the boundary between immiscible liquids, the passive perception of background noise.

- **Medium interfaceness.** Characteristic of most everyday acts of embodiment: the perception of light, reading a text, logical deduction.

- **Strong interfaceness.** Arises at points of qualitative leaps: quantum measurement, creative insight, a phase transition.

- **Extreme interfaceness.** Limiting cases where the interface reaches maximal intensity: the event horizon of a black hole, the singularity of the Big Bang (the lower bound of distinction), a moment of radical personal transformation.

The gradedness of interfaceness explains why some transitions occur "smoothly" and predictably, while others are accompanied by leap-like changes and the generation of something radically new. The intensity of the interface determines how profoundly the target-medium transforms the pattern and how emergent the result will be.

Two Aspects of Interfaceness (Inter- and Aut-)

Structurally, interfaceness can be decomposed into two complementary aspects corresponding to two fundamental functions of a boundary.

- **Inter-aspect** (from Latin *inter* — "between") characterizes the capacity of an interface to connect **qualitatively distinct ontological regimes**. This is the aspect of inter-world translation, which makes possible the embodiment of patterns from one world into another without mixing their ontological laws. The inter-aspect is responsible for preserving the autonomy of worlds while establishing a connection between them. It manifests in phenomena such as the embodiment of an idea in a material artifact ($W_2 \to W_1$), the verbalization of an emotion ($W_3 \to W_2$), translation ($W_2 \to W_2$ via W_4), and quantum measurement (W_2/mathematical formalism $\to W_1$/instrument reading).

- **Aut-aspect** (from Greek *autos* — "self," and English *outer*) fuses two related capacities of an interface. On the one hand, it points to the **inner otherness** of a single ontological regime — its capacity to differ within itself, to be non-identical to itself in different states. This is the potential for intra-world transition. On the other hand, the aut-aspect refers to the capacity of an interface to be a **point of external connectivity** — a node through which a given world joins the global network of Superreality (KSS $\to \infty$). It manifests in phase transitions (W_1), logical paradoxes (W_2), reflection and self-consciousness (W_3), and autocatalytic processes.

The distinction between the inter- and aut-aspects is not an ontological division; it is an analytical distinction that permits the description of different modes of operation of an interface. In concrete acts of embodiment, both aspects typically coexist and interact. The higher the intensity of the interface, the more pronounced both aspects become, and the more difficult it is to separate them analytically.

The Spectrum of Manifestations of Interfaceness in Various Ontological Contexts

Interfaceness manifests in all known worlds, but the specificity of its manifestation depends on the ontological flesh of the corresponding world.

- **In W_1 (the material world),** interfaceness appears as physical interactions (gravitational, electromagnetic, strong, weak); boundaries of media (surfaces, membranes,

event horizons); phase transitions and critical points; quantum measurements and wavefunction collapse.

- **In W_2 (the semantic world),** interfaceness is realized through logical operations and rules of inference; metaphors, analogies, and semantic transfers; acts of definition and the generation of new concepts; hermeneutic circles and interpretive schemes.

- **In W_3 (the phenomenal world),** interfaceness appears as thresholds of sensitivity and the discernment of qualia; associative connections and streams of consciousness; reflexive acts (the awareness of emotions, self-observation); empathy and intersubjective resonances.

- **In W_4 (the world of boundaries),** interfaceness is the very "flesh" of this world: Γ-operators of varying complexity; events of transition and translation; recursive structures of boundaries ($W_4 \to W_5 \to \ldots$).

- **In the hypothetical worlds (W_5–W_7),** interfaceness presumably manifests as the application of norms and rules (W_5); the execution of code and rendering of virtual objects (W_6); bifurcations and strange attractors in chaotic systems (W_7).

The spectrum of manifestations of interfaceness demonstrates the universality of this property: it permeates all ontological regimes, taking forms appropriate to their flesh and laws.

The Polyvalence of Interfaces

One of the key characteristics of interfaceness is **polyvalence** — the capacity of one and the same interface to link not two, but a **multitude** of ontological regimes simultaneously. This distinguishes an interface from a simple channel or bridge, which are usually conceived as paired connections.

The polyvalence of an interface manifests in that an act of embodiment can simultaneously involve several worlds. For instance, a scientific discovery may include: a

mathematical formulation (W_2); an intuitive insight (W_3); a physical experiment (W_1); a digital simulation (W_6); normative implications (W_5). All these aspects are linked by a single interface — the very act of discovery — which does not disintegrate into isolated pairwise transitions but forms an integral polyvalent event.

The polyvalence of interfaces explains why the same patterns (symmetry, resonance, feedback, cyclicity) appear in such diverse ontological contexts. A pattern is not "copied" into different worlds piecemeal; it is embodied in them **simultaneously**, through a single interface that coordinates multiple translations. The degree of polyvalence (the number of worlds linked) is one measure of the intensity of interfaceness.

Interfaceness and Scale Asymmetry (SA $\to \infty$)

Interfaceness does not exist in isolation from the axiomatics of the MPO-System; it is subject to the same fundamental principles as all of Superreality. Of particular importance is the connection between interfaceness and the axiom of scale asymmetry (SA $\to \infty$).

- **The lower bound of interfaceness.** According to SA $\to \infty$, there exists a minimal scale of distinction below which the notion of "transition" loses meaning. For interfaceness, this implies the existence of a **minimal Γ-operator** — a quantum of transition, an "atom" of interfaceness. In W_1 this threshold is the Planck length and Planck time; in W_2 — atomic logical operations; in W_3 — minimal discernible qualia.

- **Absence of an upper bound.** In the opposite direction — toward complexification and enlargement — interfaceness **has no principled limits**. Composite, recursive, hierarchically organized interfaces of arbitrarily high complexity are possible (language, scientific theory, social institution).

- **Infinite recursion of boundaries.** The absence of an upper bound on interfaceness is the ontological ground for the infinite recursion of boundaries ($W_4 \to W_5 \to \ldots$). Each new level of boundaries arises as an enlargement of the scale of distinction.

- **Directedness of interfacial transitions.** Scale asymmetry imparts directedness to interfaces: upward movement (complexification) is always possible, downward movement (simplification) is bounded by the lower limit. This explains why chains of re-embodiments exhibit a tendency toward increasing complexity, and why reverse embodiment is never an exact inversion of the forward.

Thus, interfaceness and SA $\to \infty$ stand in a relation of mutual determination: SA $\to \infty$ provides the scale bounds and directedness of interfaceness, while interfaceness is the mechanism through which scale asymmetry is operationally realized in acts of embodiment.

Chapter 14. Interfaceness and the Properties of the MPO-System

Interfaceness (P38) does not exist in isolation; it is woven into a network of relations with other properties of the MPO-System. This chapter examines the key connections of P38 with relevant properties of the system.

Relation to Property 34 "Bindability"

- **Bindability (P34)** is the operator of transition that effects the Γ-operation. It describes the capacity of a pattern and a medium to enter into a relation that generates an embodiment.

- **Interfaceness (P38)** is the ontological status of the zone in which bindability is realized. If P34 answers the question "How is the transition accomplished?", P38 answers the question "Where and in what mode does the very possibility of transition exist?"

The relation between P34 and P38 can be characterized as **tactical and strategic**. Bindability is the tactical mechanism that builds concrete connections. Interfaceness is the strategic structure that makes possible the very existence of zones in which such connections can be established. Bindability is realized *within* the interface.

Relation to Property 25 "Propertylessness" and Property 33 "Onticity"

- **Propertylessness (P25)** is the state of a pattern prior to embodiment, pure potentiality.
- **Onticity (P33)** is the status of an entity after successful embodiment, actualized being.
- **Interfaceness (P38)** is that **in which** the transition from P25 to P33 occurs. It constitutes a distinct ontological mode — the mode of the boundary. P25 and P33 are "before" and "after"; P38 is "during" (in the sense of meta-time). Without P38, the transition would be impossible; interfaceness is the **medium of embodiment**.

Relation to Property 4 "Emergence"

- **Emergence (P4)** is the generation of new properties as a result of embodiment.
- **Interfaceness (P38)** provides the **conditions** for emergence. It is precisely in the zone of translation, at the boundary of worlds, that the space for a qualitative leap arises. The interface does not merely transmit structure; it transforms it, adding to it the properties of the target-medium. Without an active boundary, emergence would be impossible. The higher the intensity of the interface, the more pronounced the emergence of the result.

Relation to Property 11 "Non-Locality" and Property 32 "Entanglement"

- **Non-Locality (P11)** and **Entanglement (P32)** describe connection without spatiotemporal contact.
- **Interfaceness (P38)** is their **ontological foundation**. It is through the interface that non-local correlation is established without signal transfer. W_4 is the site where worlds become "entangled." Quantum entanglement, semantic circles, empathy — all are manifestations of interfaceness in their respective worlds.

Relation to Property 36 "Propertyness" ($\mathcal{N}_p$) and Property 37 "Salience"

- **Propertyness (P36, $\mathcal{N}_p$)** is a quantitative measure of the complexity of the result.
- **Salience (P37)** is a measure of the intensity of the transition.
- **Interfaceness (P38)** determines the gradedness of both these properties. The intensity of the interface correlates with the threshold of salience and the increment in $\mathcal{N}_p$. The relation among P38, P37, and P36 can be represented as a three-dimensional coordinate system that defines the "ontological energetics" of the transition.

Chapter 15. The Auto-Interface and Intra-World Interfaceness

The Concept of the Auto-Interface

The **auto-interface** is a limiting case of interfaceness, in which the transition occurs **within a single ontological regime**. It is not a boundary between worlds but a boundary between different states, patterns, or scales of one and the same ontological medium. In the auto-interface, the inter-aspect is reduced, and the aut-aspect comes to the fore. The auto-interface demonstrates that a world is not homogeneous and static; it contains internal distinctions and potentials for self-transformation. Formally, the auto-interface obeys all the laws of interfaceness (P38): it is graded, can be polyvalent, is subject to SA $\to \infty$, and is realized through Γ-operators.

Examples of Auto-Interfaces in W_1, W_2, W_3

- **In W_1:** a phase transition (water $\to$ ice), a quantum leap, a bifurcation in a nonlinear system.
- **In W_2:** a logical paradox ("This statement is false"), the moment of proving a theorem, the definition of a new concept.
- **In W_3:** reflection (awareness of an emotion), insight, personal transformation.

The Status of the Auto-Interface in the MPO-System

The auto-interface performs several important functions:

- It confirms the **totality of PE**: PE operates not only between worlds but also within them.
- It refutes the view of PE as an exclusively inter-ontological phenomenon.
- It provides a **unified language** for describing intra-world dynamics.
- It links the universality of PE with concrete manifestations of dynamics in each of the worlds.

Part VI. PE, Scale Asymmetry, and Evolution Without Darwinism: From Ontology to Empirical Implications

Chapter 16. Scale Asymmetry (SA $\to \infty$) as a Condition for the Directedness of PE

The axiom of scale asymmetry (SA $\to \infty$) was introduced in Chapter 2 as one of the four fundamental postulates of the MPO-System. Its main implications for the Phenomenon of Embodiment were examined in Chapter 11. In the present chapter, the analysis is deepened: SA $\to \infty$ is considered as an ontological condition that imparts **directedness** to chains of re-embodiment and, ultimately, serves as the foundation for evolution as a necessary process.

The Lower Bound of Distinction

SA $\to \infty$ asserts that in each ontological world there exists a **minimal scale of distinction** below which the notion of "embodiment" loses meaning. This bound has an **ontological status** — that is, it belongs to the very structure of reality.

- **In W_1:** the Planck length ($\sim 1.6 \times 10^{-35}$ m), Planck time, Planck mass.

- **In W_2:** atomic concepts that cannot be further decomposed without loss of meaning (logical constants, basic categories).
- **In W_3:** thresholds of sensitivity below which experience becomes indiscernible.

The lower bound imparts **finitude** to each concrete act of embodiment: a pattern cannot be embodied arbitrarily "finely."

The Absence of an Upper Bound

In the opposite direction — toward enlargement, complexification, integration — SA $\to \infty$ **imposes no principled limit whatsoever**. A pattern can be embodied in ever more complex and large-scale structures without restriction (from quarks to galaxies, from atomic concepts to meta-theories, from the simplest sensations to collective forms of experience). This means that the chain of re-embodiments **is never obliged to stop**.

Irreversibility and Uniqueness

From scale asymmetry there directly follows the **irreversibility** of chains of re-embodiment and the **uniqueness** of each act.

- **Irreversibility:** Exact reverse embodiment is impossible in principle, because downward movement on the scale is bounded, while upward movement irreversibly enriches the pattern with new emergent properties that cannot be "accommodated" at a lower scale.
- **Uniqueness:** Each act of PE generates a unique entity. In the chain of re-embodiments, there are no two identical states.

This provides an ontological grounding for the **arrow of time** and the historicity of being.

SA $\to \infty$ and the Structure of the World of Boundaries (W_4)

Scale asymmetry applies to the world of boundaries W_4 itself. This means:

- There exists a **minimal Γ-operator** — a quantum of transition.
- There is **no upper bound** on the complexity of Γ-operators.
- The infinite recursion of boundaries ($W_4 \to W_5 \to \ldots$) is a direct consequence of the absence of an upper bound.

These properties guarantee that the process of embodiment never hits a "final" interface.

PE as the Operational Expression of SA $\to \infty$

SA $\to \infty$ is realized in each act of embodiment through:

- The **choice of scale of the target-medium** (bounded below, open above).
- The **determination of the threshold of salience (P37)** (scale-dependent).
- The **establishment of the propertyness ($\mathcal{N}_p$) of the result** (can grow without limit).

Thus, PE is the very way in which scale asymmetry **acts** in Superreality, manifesting itself in the directedness of technological progress, biological evolution, and cosmological expansion.

Implications for "Hard Problems"

- **Consciousness:** arises upon exceeding the lower threshold of $\mathcal{N}_p \approx 10^6$.
- **Cosmology:** the Big Bang as the attainment of the lower bound; expansion as upward movement.
- **AI:** the possibility of strong AI is linked to the attainment of critical $\mathcal{N}_p$ and interfaceness (P38).

Chapter 17. Evolution as Necessary Complexification: A Model Without Darwinism

Critique of the Darwinian Mechanism as Insufficient

The Darwinian mechanism (random variation + selection) explains **adaptation** within a level but not the **transition** to a new level of complexity. It does not explain the directedness of evolution, which is observed not only in biology but also in technology, science, and culture.

PE + SA $\to \infty$ as the Ontological Basis of Evolution

Key elements of the model:

1. Each embodiment is unique and enriches the pattern (P4).
2. The result (P33) becomes the pattern (P25) for the next act.
3. SA $\to \infty$ imparts directedness: downward movement is bounded, upward movement is open.
4. The selection of stable embodiments occurs through the attainment of a threshold of salience (P37) and stabilization via bindability (P34).

Evolution is a **directed process of actualization of the potential**, built into the architecture of Superreality. Directedness does not imply a goal.

Rejecting Teleology Without Rejecting Directedness

The model affirms **directedness** (growth of $\mathcal{N}_p$) but denies **teleology** (a pre-given goal). Evolution is contingent in detail but non-random in its overall vector.

Empirical Implications of the Model

1. **Thresholds of complexity:** transitions should occur upon reaching critical $\mathcal{N}_p$ (for consciousness $\mathcal{N}_p \approx 10^6$).

2. **Irreversibility:** evolution does not "roll back."

3. **Universality of the trend:** complexification in biology, technology, science, culture.

4. **Acceleration:** each new embodiment accelerates further evolution.

Comparison with Darwinism

Darwinism is **not abolished but included as a special case** — a local mechanism of microevolution within W_1. The vector of macroevolution is set by SA $\to \infty$.

Philosophical Implications

- The rejection of ontological randomness.
- The restoration of the idea of **cosmic productivity**.
- An ethical dimension: the value of life lies in participation in chains of embodiment.
- A new view of the place of the human.

Chapter 18. Evolution as Ontologically Necessary Complexification: The Synthesis of PE and SA $\to \infty$

Note: Chapter 17 contains the argumentation of the model; Chapter 18 focuses on the synthesis and additional aspects.

Rejecting Randomness as an Ontological Principle

Randomness in the MPO-System is an **epistemic** category (a consequence of incomplete knowledge of the Γ-operator), not an **ontological** given. The unpredictability of the future is rooted in the infinite complexity of the network of connectivity and the freedom of interfaces, not in fundamental randomness.

Empirical Domains Confirming the Model

- **Biological evolution:** leap-like transitions (the Cambrian explosion, etc.).
- **Technological development:** exponential growth of complexity.
- **Scientific cognition:** scientific revolutions as moments of extreme interfaceness.
- **Cultural evolution:** the complexification of symbolic systems and institutions.

The Status of Darwinism in the New Framework

Darwinism retains its power for microevolution within a level but is not a fundamental ontological principle.

Predictions and Empirical Implications

The predictions are summarized and extended: thresholds of complexity, irreversibility, acceleration, convergence, and the interfaceness of AI.

Philosophical Implications

- **Ontological optimism:** being is productive.
- **Freedom and responsibility:** an ontological space for freedom and irreversible consequences.
- **Meaning and value:** meaning is generated in the process of embodiment through the growth of $\mathcal{N}_p$.

Chapter 19. Empirical Implications of the Model of the Phenomenon of Embodiment

The model generates empirically testable hypotheses. These implications stem from three components: (1) SA $\to \infty$, (2) the metric $\mathcal{N}_p$, (3) the schema of PE as a chain of re-embodiments.

Diagnosing Thresholds of Complexity Through Propertyness ($\mathcal{N}_p$)

As was shown in Chapters 16–18, scale asymmetry and chains of re-embodiments impose the ontological necessity of directed complexification. The quantitative measure of this complexification is propertyness ($\mathcal{N}_p$).

- **Hypothesis:** The threshold of consciousness corresponds to $\mathcal{N}_p \approx 10^6$.
- **Empirical tests:** analysis of neural networks (AI), investigation of biological systems, analysis of social structures. Prediction: for $\mathcal{N}_p < 10^6$, reflexive capacities are absent; for $\mathcal{N}_p \geq 10^6$, they appear leap-like.

Testing Interfaceness (P38) in Artificial Intelligence

The analysis conducted in Part V showed that interfaceness (P38) is the key property that enables the translation of patterns across ontological regimes.

- **Hypothesis:** True intelligence manifests in the capacity to translate structures across worlds (data $\rightarrow$ metaphor, fact $\rightarrow$ analogy).
- **Test battery:**
  1. **Test of metaphorical transfer** (physical process $\rightarrow$ social/emotional sphere).
  2. **Test of feedback** (correction of a plan based on the result of an action).
  3. **Test of polyvalence** (simultaneous interpretation of a pattern in W_1, W_2, W_3).

Analyzing Chains of Re-Embodiment in the History of Technology and Science

In Chapters 17–18, it was shown that evolution in a broad sense can be described as a chain of re-embodiments directed by scale asymmetry.

- **Methodology of reconstruction:** idea (W_2) $\rightarrow$ blueprint/plan (W_2/W_4) $\rightarrow$ artifact (W_1) $\rightarrow$ experience (W_3) $\rightarrow$ new idea (W_2'). The preservation of the invariant, emergent enrichment, and growth of $\mathcal{N}_p$ are recorded.

- **Historical cases:** the evolution of computing devices (abacus → cloud computing), the development of the theory of gravitation (Aristotle → quantum gravity).
- **Prognostic value:** prediction of leaps to a new level of complexity at points of maximal salience (P37).
- **Rejection of teleology:** the chain does not move toward an "ideal" but develops due to the ontological necessity of SA $\to \infty$.
- **Empirical testability:** formalization through the metrics $\mathcal{N}_p$ and P37, comparison with historical data.

Part VII. Comparative Analysis with Other Ontologies

Chapter 20. The Place of the MPO-System in the Contemporary Philosophical Landscape

The Necessity of Comparative Analysis

Comparative analysis permits a refinement of the specificity of the MPO-System, a demonstration of its continuity, and a test of the system's robustness. Three relevant traditions are considered: the process philosophy of Whitehead, the actor-network theory of Latour, and the object-oriented ontology of Harman.

Criteria of Comparison

1. The fundamental unit of ontology.
2. The status of relations.
3. The plurality of ontological regimes.
4. The role of potentiality.
5. The presence of scale asymmetry.
6. A quantitative metric of complexity.

Chapter 21. The Process Philosophy of A.N. Whitehead

Basic concepts. In his major work *Process and Reality* (1929), Alfred North Whitehead constructs a cosmology in which the fundamental units are not stable substances but **actual entities** — moments of experience that "prehend" other entities and unite into **nexūs**. **Creativity** is the universal principle of transition from possibility to actuality. **Eternal objects** are pure potentiality, analogous to Platonic ideas, but actualized only in concrete events. The world, for Whitehead, is a process of becoming, where "being" is always a possibility for "becoming." **God** in Whitehead's system functions as the "principle of concretion" — holding eternal objects and securing their ingression into actual entities.

Proximity to the MPO-System. Both systems share a radical **anti-substantialism**: reality is understood as process rather than as a collection of things. Whitehead's **eternal objects** are close to the MPO-System's **patterns (P25)** — pure potentiality awaiting actualization. **Creativity** resonates with the **Γ-operator** as the mechanism of transition from the possible to the actual. The very idea of a **nexus** echoes **interfaceness (P38)** and the axiom **KSS** $\to \infty$ (absolute connectivity). Finally, both systems reject reductionism and acknowledge a plurality of levels of reality, though they interpret this differently.

Differences. Despite their deep kinship, there are fundamental divergences between the systems.

1. **Pluralism of worlds.** The MPO-System postulates a **multitude of ontologically heterogeneous worlds** (W_1–W_n) with their own "flesh" and laws, irreducible to one another. Whitehead constructs a **unified processual cosmology**: all actual entities belong to a single ontological order, though they differ in degree of complexity and "strength" of prehension. For the MPO-System, worlds differ **qualitatively** (matter vs. meaning vs. experience); for Whitehead, they differ **quantitatively** (degree of organization of experience).

2. **Scale asymmetry.** Whitehead formulates no analogue of the axiom **SA** $\to \infty$. In

his cosmology, the scale (from the simplest actual entities to complex societies) is symmetrical or, at any rate, not thematized as asymmetric. The notion of a lower bound on distinction and the consequent irreversibility are absent.

3. **The role of God.** Whitehead retains **God** as a necessary element of the system — the "principle of concretion" without which eternal objects could not enter into actual entities. For Whitehead, God performs a dual function: holding eternal objects in their potentiality ("primordial nature") and securing their harmonious ingression into the world ("consequent nature"). Without this agent, the processual cosmology lacks completion: someone is needed who "prehends" all possibilities and co-determines their actualization.

   At first glance, the MPO-System replaces this function with an impersonal axiomatics: **KSS** $\to \infty$ guarantees connectivity, **PPU** $\to \infty$ guarantees the stability of paradoxes, **SA** $\to \infty$ guarantees directedness. No agent or privileged entity is required to perform these functions; they are built into the very architecture of Superreality.

   Yet such a view would be incomplete. The conceptual inexhaustibility of Superreality — the infinite recursion of boundaries ($W_4 \to W_5 \to \ldots$), the absence of an upper scale bound (SA $\to \infty$), the principled locality of all cognition (the "epsilon-neighborhood") — indicates that in human (and indeed any finite) cognition of reality, there is inevitably a domain that cannot be exhausted by any formalization or natural-scientific investigation. This is not a temporary gap to be filled by future science, but a **structural boundary** imposed by the very architecture of Superreality.

   Historically, this domain has been marked by the name **"God"** — not as the name of a concrete anthropomorphic agent, but as an indication of that "with which we must reckon, although we do not and cannot know what it is." This is the domain where the possibilities of the natural sciences come to an end and the necessity of purely metaphysical (or theological) work begins: we are compelled somehow to

make sense of and to take account of this boundary, without claiming to know it definitively.

Conscientious and profound investigators — from Plato and Aristotle to Kant, Whitehead, and contemporary cosmologists — have inevitably come to acknowledge such a domain. Only intellectual charlatans can claim that their conception **excludes** the idea of God as such or, on the contrary, **fully defines** it. The MPO-System claims neither. It does not postulate God as an entity, nor does it eliminate the very problematic connected with this domain. Instead, it **fixes a structural place** for what in various traditions has been named God: that place is universal interfaceness (P38) as the ultimate horizon of connectivity, paradoxial stability, and directedness, which always eludes definitive objectification but with which any serious ontological construction must reckon.

Thus, the difference between Whitehead and the MPO-System on the question of God is not reducible to a simple "replacement" of an agent by axioms. Whitehead introduces God as an **intrasystemic agent** performing concrete functions. The MPO-System distributes these functions among the axioms, yet acknowledges that the very fullness and inexhaustibility of Superreality points to a horizon that functions in human cognition as "God" — not as an object, but as the condition of possibility of all objectification and simultaneously its limit. This is not theism in the traditional sense, but neither is it atheism; it is **ontological honesty** concerning the fact that finite cognition always encounters the infinite, and this encounter demands its own — metaphysical or theological — language.

4. **Quantitative metric.** Whitehead offers no formal measure of complexity. The MPO-System introduces **propertyness** ($\mathcal{N}_p$), which permits (at least hypothetically) the comparison of the ontological complexity of different entities and the setting of thresholds for qualitative transitions.

5. **Interfaceness.** The notions of W_4 as an autonomous world of boundaries and **P38** as a property of an active zone of translation have no direct analogue in Whitehead.

Although prehension can be interpreted as a proto-interface, Whitehead does not single out the boundary as a distinct ontological regime.

Thus, the MPO-System inherits Whitehead's processuality and anti-substantialism, but radicalizes them through (1) the pluralization of ontological regimes, (2) the introduction of scale asymmetry, (3) the elimination of the theological remainder, and (4) the development of a formal apparatus (properties, metrics).

Chapter 22. The Actor-Network Theory of B. Latour

Basic concepts. Bruno Latour, one of the founders of actor-network theory (ANT), proposes a **"flat ontology"** in which humans and non-human entities (artifacts, animals, technologies, natural objects) are regarded as equipollent **actors** in **hybrid networks**. Agency is not an intrinsic property of a subject or object; it is **distributed** and depends on embeddedness in a network of relations. Objects are understood as **"institutions"** — stable assemblages that delegate action and make possible the stable functioning of sociotechnical systems. Latour radically rejects the binary oppositions of modernity (nature/culture, subject/object, fact/value) as artificial constructs requiring "purification work."

Proximity to the MPO-System. The MPO-System shares several fundamental stances with ANT:

- **The rejection of binary oppositions.** The distinction of worlds in the MPO-System (W_1, W_2, W_3, W_4) is not a reproduction of the "matter/consciousness" or "nature/culture" opposition. Worlds are not opposed but juxtaposed; transitions between them are accomplished through interfaces.

- **The primacy of connections.** The axiom **KSS** $\to \infty$ (absolute connectivity) is the ontological version of Latour's thesis of the universal connectivity of actors. In both systems, there are no isolated entities; everything exists within a network.

- **The decentering of the human.** The MPO-System, like ANT, does not accord the human a privileged ontological status. Human consciousness (W_3) is one onto-

logical regime alongside the material (W_1) and the semantic (W_2), not the "crown of creation" or a unique source of meaning.

- **Interfaceness and "institutions."** The principle of **interfaceness (P38)** is close to Latour's understanding of an object as an **"institution"** — a stable assemblage linking different orders of reality and delegating action.

Differences. Despite the considerable convergence of stances, there are principled divergences between the MPO-System and ANT.

1. **Flatness vs. stratification.** Latour insists on a **"flat ontology"**: there are no levels, no hierarchies; everything exists on a single plane of the network. The MPO-System, by contrast, acknowledges **qualitatively distinct ontological regimes** (ChOR $\to \infty$). Worlds are not hierarchical (there are no "higher" or "lower"), yet they are **heterogeneous** in their flesh and laws. This is not a flat ontology but a **topology of heterogeneous layers** linked by interfaces.

2. **Potentiality.** ANT lacks a developed notion of pure potentiality. Actors are always already given in the network; possibility is not thematized as an independent mode of being. The MPO-System, on the contrary, assigns a central role to **Propertylessness (P25)** — the state of a pattern prior to embodiment. PE is the transition from potentiality to actuality, and this transition constitutes the core of Superreality's dynamics.

3. **Quantitative metric and axiomatics.** ANT remains descriptive and empirical; it offers no formal axiomatics or quantitative measures of complexity. The MPO-System, as an ontological system, introduces axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$, SA $\to \infty$) and metrics ($\mathcal{N}_p$, P37), which permit not only description but also modeling and prediction.

4. **The world of boundaries.** The notion of W_4 as an autonomous ontological regime whose "flesh" consists of events of transition has no analogue in ANT. For Latour, a boundary is always a boundary between actors in a network, not a distinct type of entity.

5. **Directedness and evolution.** ANT offers no model of directed evolution or complexification. Networks can be reassembled, but there is no built-in vector. The MPO-System, through **SA** $\to \infty$, introduces directedness as an ontological necessity.

Thus, the MPO-System borrows from ANT the primacy of connections and the decentering of the human, but places these intuitions within a more structured ontological framework, with a distinction of worlds, formal axiomatics, and directed dynamics.

Chapter 23. The Object-Oriented Ontology of G. Harman

Basic concepts. Graham Harman, the leading figure of object-oriented ontology (OOO), asserts that **everything is an object** — from a mailbox to electromagnetic radiation and fictional entities. Objects possess **autonomy** and a **"withdrawn surplus of being"** that cannot be reduced either to their parts (undermining) or to their relations with other objects (overmining). Any object exists **outside all relations**; relations merely "translate" or "distort" objects but never exhaust them. OOO radically criticizes **"correlationism"** — the stance that reduces reality to the correlation between the human and the world (access to being only through thought or language).

Proximity to the MPO-System. Both systems are metaphysical projects that claim universality and reject anthropocentrism. Both reject correlationism: reality is not reducible to how it is given to the human subject. The principle of **ontological egalitarianism** in OOO (all objects are equally real) echoes **ChOR** $\to \infty$: worlds are not hierarchical, none is privileged, all ontological regimes are equipollent.

Differences. Despite the shared anti-correlationist pathos, the MPO-System and OOO diverge radically on key questions.

1. **The status of relations.** This is perhaps the deepest divergence. The MPO-System is **radically relational**: being is a node of connections, and there are no objects outside relations. An object in the MPO-System is a secondary stabilization within the field of connectivity, not an autonomous entity with a "withdrawn

surplus." For Harman, by contrast, objects possess an irreducible inner core that **principally eludes** all relations. Relations do not constitute the object but merely "touch" its surface. The MPO-System is thus an **ultra-relationalism**, whereas OOO is an **anti-relationalism**.

2. **Processuality vs. staticity.** The MPO-System is a processual ontology: the Phenomenon of Embodiment is the universal mechanism of becoming. OOO is often criticized for a **static substantialization** of objects. Although Harman speaks of "four-dimensional objects" and events, the general tenor of his philosophy is of objects as stable, self-identical entities whose change is problematic.

3. **The Γ-operator and the act of embodiment.** OOO has no analogue of the Γ-operator as an event of transition between ontological regimes. The interaction of objects is described through the notion of "causality" as "vicarious causation" or "translation," but without a detailed mechanism comparable to PE.

4. **Plurality of worlds.** Although OOO acknowledges an infinite diversity of objects, it does not systematically distinguish **ontological regimes** with their own flesh and laws. For Harman, all objects exist on a single "plane" (though with varying depths of withdrawal). The MPO-System distinguishes qualitatively different types of existence (matter, meaning, experience, boundary), which constitutes a more differentiated ontological architecture.

5. **Quantitative metric and scale asymmetry.** OOO offers no quantitative measures of complexity and does not thematize scale asymmetry. The notions of $\mathcal{N}_p$, SA $\to \infty$, thresholds of salience are absent.

Thus, the MPO-System shares with OOO the anti-correlationism and egalitarianism, but realizes them in a radically relational, processual, and structurally differentiated form. Where OOO postulates autonomous, withdrawn objects, the MPO-System sees nodes of connectivity and interfaces. Where OOO struggles to explain change and interaction, the MPO-System offers the Phenomenon of Embodiment.

Chapter 24. Unique Components of the MPO-System

The comparative analysis permits the identification of several components that together constitute the uniqueness of the MPO-System within the contemporary philosophical landscape. None of these components individually is absolutely new (analogues and anticipations can be found), but their **systematic integration** within a unified axiomatics and operational phenomenology is unprecedented.

1. **The integration of absolute connectivity (KSS $\to \infty$) and scale asymmetry (SA $\to \infty$).** None of the systems examined combines the ontological postulate of universal connectivity with the principle of a directed scale.

2. **Property 38 "Interfaceness."** The notion of a boundary as an active zone of translation, with graded intensity, polyvalence, and the distinction of inter- and aut-aspects, has no direct analogues.

3. **A quantitative metric of ontological complexity ($\mathcal{N}_p$).** The introduction of propertyness as a measure of ontological complexity, which permits (in prospect) operationalization and empirical testing, takes the MPO-System beyond purely descriptive metaphysics.

4. **A clear distinction of ontological worlds (W_1–W_n).** In contrast to "flat" ontologies and monistic cosmologies, the MPO-System offers a pluralistic architecture with qualitatively distinct regimes.

5. **The Phenomenon of Embodiment (PE) as a universal processual mechanism.** The MPO-System offers a formal structure for the act of embodiment (P25 $\to \Gamma \to$ P33) and links it to the axiomatics and properties.

6. **The rejection of a "theory of everything" and reflexive honesty.** The MPO-System fixes the principled inexhaustibility of Superreality and the locality of all cognition (the "epsilon-neighborhood").

Taken together, these components constitute what may be termed a **meta-ontology**:

not just another theory of being, but a framework within which various ontological regimes and the transitions between them can be mapped and analyzed.

Part VIII. Conclusion: Superreality as a Field of Embodiment

Chapter 25. Summary of Results

The Phenomenon of Embodiment as the Processual Core of Superreality

PE is the universal mechanism for the actualization of the potential (P25 $\to$ P33), operating totally (both between worlds and within them). Its key characteristics: the invariance of the pattern, the rejection of transfer, cyclicity/irreversibility, and a three-phase structure.

The Axiomatic Foundation

The four axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$, SA $\to \infty$) form an indissoluble unity that provides ontological diversity, connectivity, stability under paradoxes, and directedness.

Interfaceness (P38) as the Completing Property

P38 fixes the ontological status of the boundary as an active zone of translation. It is graded, polyvalent, distinguished by aspects, and subject to SA $\to \infty$. It is a full-fledged property that secures the operational connection between the axiomatics and the acts of PE.

Empirical Significance

The model generates testable hypotheses: thresholds of complexity ($\mathcal{N}_p \approx 10^6$ for consciousness), tests of interfaceness for AI, and the analysis of historical chains of re-embodiment.

Chapter 26. Philosophical and Scientific Implications

Overcoming Reductionism

The MPO-System overcomes physicalism, idealism, and other forms of reductionism by showing that they are category mistakes. Consciousness and brain are two interfaces of one node of connectivity; meaning is not an epiphenomenon; causality is replaced by isomorphism.

Evolution Without Darwinism

Directed complexification is an ontological necessity. Darwinism is included as a special case of microevolution.

A New View of "Hard Problems"

- **Consciousness:** the question of the threshold of $\mathcal{N}_p$.
- **Quantum gravity:** the problem of the lower bound of distinction in W_1.
- **The nature of time:** the arrow of time is grounded by SA $\to \infty$.

Rejecting a "Theory of Everything"

The MPO-System fixes the principled inexhaustibility of Superreality and the locality of cognition, offering not final answers but a **map and a method**.

Chapter 27. Open Questions and Directions for Further Research

1. **Formalization of the Γ-operator** (category theory, a metric on the space of worlds, 2-categories).
2. **Operationalization of the metric $\mathcal{N}_p$** (algorithms for computation, units of measurement, calibration).
3. **Experimental verification of thresholds** (tests on AI, neurobiological data, historical reconstructions).

4. **Expansion of the map of worlds** (phenomenological investigation of new candidates).

5. **Dialogue with theology** (making sense of the structural isomorphism between the axiomatics of the MPO-System and the function of God).

Appendices

Appendix A. Schema of the Phenomenon of Embodiment (Operational Model)

The present schema captures the structural logic of the Phenomenon of Embodiment as the processual core of Superreality. It integrates the axiomatics, key properties (above all P38 "Interfaceness"), and the dynamics of re-embodiment, excluding reductionist metaphors ("transfer," "energy," "substance").

A.1. Architecture of the Schema

The schema is built on three levels:

1. **The ontological foundation** — the axioms and the field of connectivity.

2. **The operational contour** — the mechanism of the act of embodiment through the interface.

3. **Dynamic topology** — chains of re-embodiment and scale asymmetry.

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┌──────────────────────────────────────────────────────────────────────────────────┐
│                    SUPERREALITY: THE FIELD OF ABSOLUTE CONNECTIVITY (KSS → )     │
│        Axioms: ChOR →  | KSS → | PPU →  | SA →                                  │
└──────────────────────────────────────────────────────────────────────────────────┘
```

THE OPERATIONAL CONTOUR OF PE

SOURCE
(ChOR)

Pattern
State:
P25
(Propertyless-
ness)

THE WORLD OF BOUNDARIES (W)
INTERFACE / -OPERATOR

Property 38: INTERFACENESS (P38)

- Actualization of connectivity (KSS)
- Establishment of isomorphism
- Polyvalence / Transformation
- Event in meta-time
- Bindability (P34) as tactics

TARGET
(ChOR)

Medium

Invariant of structure
(Isomorphism without transfer)

Emergence (P4)
Salience (P37)

RESULT: THE EMBODIED ENTITY

- Status: P33 (Onticity)
- Medium: Authentic for ChOR
- Properties: Stability (P7), Integrity (P2), Propertyness ()
- Function: Becomes a new Pattern for the next act (P25')

```
                    DYNAMIC TOPOLOGY: THE CHAIN OF RE-EMBODIMENT

... → [P25 →  → P33] → [P25' → ' → P33'] → [P25'' → '' → P33''] → ...

• Directedness: Scale Asymmetry (SA → )
  - Floor: The lower bound of distinction (minimal scale of )
  - Ceiling: Absence of a limit (infinite complexification,  → )
• Irreversibility: Exact reverse embodiment is impossible (uniqueness of P33)
• Cyclicity: A network without beginning or end; any P33 is P25 for the next
• Locality/Globality: Each act is local but rooted in KSS →  via P38
```

A.2. Key Blocks and Connections

- **Source (ChOR$_i$):** The pattern in state P25, providing an invariant structure.
- **The World of Boundaries (W_4) and the Γ-Operator:** W_4 is the regime of events of transition; the Γ-operator is the event-interface that actualizes connectivity through P38.
- **Target (ChOR$_j$) and Medium:** Actively shapes the result through its laws.
- **Result: The Embodied Entity:** Status P33, characterized by isomorphism, emergence (P4), and propertyness ($\mathcal{N}_p$).

A.3. Practical Application of the Schema

1. **Analysis of phenomena:** decomposition into pattern, Γ-operator, medium, and result.
2. **Design of interfaces:** emphasis on Γ-operators as zones of translation.

3. **Diagnostics of thresholds:** use of $\mathcal{N}_p$ and SA $\to \infty$.

4. **Avoidance of reductions:** blocking the reduction of one world to another.

Appendix B. Glossary of the MPO-System: The Phenomenon of Embodiment and Superreality

Version 1.0 (based on the monograph "The Phenomenon of Embodiment: The Evolution of Ontology from Properties to Interfaces")

B.1. Fundamental Concepts and Axioms

- **Superreality:** The unified, dynamic field of absolute connectivity (KSS $\to \infty$). Objects and worlds are secondary stabilizations.
- **The axiom of infinite stratification (ChOR $\to \infty$):** Superreality consists of an infinite number of autonomous, irreducible ontological regimes.
- **The axiom of absolute connectivity (KSS $\to \infty$):** All entities across all worlds are absolutely interconnected.
- **The axiom of paradoxial stability (PPU $\to \infty$):** Superreality is stable under contradictions and paradoxes.
- **The axiom of scale asymmetry (SA $\to \infty$):** The scale of distinction is asymmetric: there is a lower bound but no upper bound.

B.2. Ontological Regimes (Worlds)

- **World (ChOR):** An autonomous medium with a unique flesh and laws.
- W_1 — **The Material World:** Flesh: mass-energy, fields.
- W_2 — **The Semantic World:** Flesh: meanings, signs, relations.
- W_3 — **The Phenomenal World:** Flesh: experiences, qualia.

- W_4 — **The World of Boundaries (Interfaces):** Flesh: the event of transition, Γ-operators.
- **Authentic medium:** The medium in which an entity exists naturally, in accordance with its laws.

B.3. The Structure of the Phenomenon of Embodiment

- **The Phenomenon of Embodiment (PE):** The fundamental process of the actualization of a pattern through the Γ-operator, generating a new entity. It transfers neither matter, nor energy, nor information; it establishes an isomorphism.
- **Re-embodiment:** A refined term that underscores the absence of a first or last act.
- **Pattern:** A stable configuration of connections in a state of pure potentiality (P25).
- **Medium (Ontological flesh):** The context of embodiment that actively shapes the result.
- **Γ-operator:** The event-interface that initiates the act of embodiment.
- **Isomorphism:** Structural correspondence, the basis of connection without substrate transfer.
- **Chain of re-embodiments:** An infinite, asymmetric sequence of acts of PE.

B.4. Properties and Metrics

- **Interfaceness (Property 38, P38):** The capacity to function as a dynamic zone of translation. Graded, polyvalent; inter- and aut-aspects are distinguished.
- **Auto-interface:** A transition within a single ontological regime.
- **Propertylessness (P25):** The pure potentiality of a pattern prior to embodiment.
- **Onticity (P33):** The status of an entity after successful embodiment.

- **Bindability (P34):** The operator of transition that effects the Γ-operation.
- **Propertyness (P36, $\mathcal{N}_p$):** A quantitative measure of ontological complexity.
- **Salience (P37):** The ontological trigger, a measure of the intensity of transition.
- **Emergence (P4):** The generation of new properties in embodiment.
- **Capacity (P35):** The ability to contain an infinite number of embodiments.

B.5. Derivative and Operational Concepts

- **Image:** A stable, plastic configuration; an interface (P38) in action.
- **Onto-plasticity:** The capacity of an image to be repeatedly re-embodied (an emergent manifestation of P38).
- **Meta-time:** The order of distinctions "before" and "after," lacking duration.
- **Across worlds:** The correct description of the direction of PE (through W_4, without violating autonomy).

Appendix C. A Map of Ontological Worlds

The table presents the key ontological regimes (worlds) of the MPO-System, their characteristics, and typical Γ-operators. The map remains open in principle for additions.

World	Name	Ontological Flesh	Types of Entities
W_1	Material	Mass-energy, fields, space-time	Particles, bodies, radiation, w
W_2	Semantic	Relation, meaning, invariant	Concepts, symbols, theorems,
W_3	Phenomenal	Experience, feeling, qualia	Sensations, emotions, images
W_4	World of Boundaries	Event of transition	Γ-operators, interfaces, acts of em
W_5	Normative (hypoth.)	Imperative, norm	Laws, rules, obligations
W_6	Virtual (hypoth.)	Code, memory state	Digital objects, simulations, N
W_7	Chaotic (hypoth.)	Dynamic instability, attractor	Turbulence, fractals, market fluc

Note. W_8 (Spiritual world) is mentioned as an unverified candidate but is not included in the main map due to the absence of reproducible data and the criteria of the MPO-System.

Primary Texts of Ontology Lab Used as a Basis for the Monograph

(available in open sources)

- A-living-thought-interface-for-human-AI-cognitive-collaboration
- MPO-S
- Superreality and Its Worlds
- Dialectics-of-Bindability
- Laboratory-of-Ontologies-MSO-S-Multiverse-of-Cognition-and-Creation-of-the-Supermasterpie
- Meet-the-37-Salience
- Meet-the-38-Interfaceness
- Topology-of-the-Limit
- CONCEPT of Dynamic Realism: The Genesis of the Property Method and Superreality
- Scale Asymmetry: The Fourth Axiom of Superreality
- Interface-and-Bindability
- Γ-in-the-Wild-An-Ontological-Gateway-for-AI-Science
- HOME: Whispering Pictures of (Phenomenology of Embodiment)
- The Eye of Superreality: How Nature Sees Itself and Stumbles Upon Its Own Blind Spot
- Scale, Asymmetry, and the Structure of Distinction (full version)

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